Starspots let us time a star's rotation, but the measured period shifts month to month. That scatter is called the wander. Different latitudes rotate at different rates: faster at equator than its poles. That difference is called shear. Fully convective M dwarfs have no radiative core. Theory predicts weaker shear in fully convective M dwarfs than in Sun-like stars, yet observations disagree. Doppler imaging suggests a steep decline toward cooler stars while Kepler’s two-period method finds a nearly flat trend.

This research used 450 stars from 3,000 to 6,000 K with common brightness, period, and sector-count limits. Periods were refitted in 4,317 sector light curves. A perfectly steady synthetic signal was injected into each star’s residual noise and re-measured, giving a star-specific noise floor that was subtracted.

Sun-like stars above 5,000 K wandered about four times more than fully convective mid-M dwarfs below 3,400 K. A cutoff of 0.015 radians per day, twice the noise floor was applied. Among Sun-like stars, 83% exceeded it. Among mid-M dwarfs, 18% did. No detectable jump appeared at the fully convective boundary. A preregistered 422-star sample reproduced the trend. Independently processed light curves showed the same deviations.

Spot-evolution simulations reproduced the Sun-like stars' wander with or without published shear. For mid-M dwarfs they predicted 2.4 times too much. The wander therefore reflects how fast spot patterns change, more than shear. Cooler stars keep better time because their spot patterns stay more coherent month to month.